



10337 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
WR48a				
	1	WR48a-O1	MIRI Medium Resolution Spectroscopy	(5) WR48a-OFFSET1
	2	WR48a-O2	MIRI Medium Resolution Spectroscopy	(6) WR48a-OFFSET2
	5	WR48a-Bkgd	MIRI Medium Resolution Spectroscopy	(7) WR48a-Bkgd
	4	WR48a-Imaging	MIRI Imaging	(3) WR48a
WR137				
	6	WR137-O1	MIRI Medium Resolution Spectroscopy	(8) WR137-OFFSET1
	7	WR137-O2	MIRI Medium Resolution Spectroscopy	(9) WR137-OFFSET2
	8	WR137-Bkgd	MIRI Medium Resolution Spectroscopy	(10) WR137-Bkgd
	9	WR137-Imaging	MIRI Imaging	(1) WR137
WR112				
	11	WR112-O1	MIRI Medium Resolution Spectroscopy	(11) WR112-OFFSET1
	12	WR112-O2	MIRI Medium Resolution Spectroscopy	(12) WR112-OFFSET2
	13	WR112-Bkgd	MIRI Medium Resolution Spectroscopy	(13) WR112-Bkgd
	14	WR112-Imaging	MIRI Imaging	(2) WR112
WR125				
	16	WR125-O1	MIRI Medium Resolution Spectroscopy	(14) WR125-OFFSET1
	17	WR125-O2	MIRI Medium Resolution Spectroscopy	(15) WR125-OFFSET2
	18	WR125-Bkgd	MIRI Medium Resolution Spectroscopy	(16) WR125-Bkgd
	19	WR125-Imaging	MIRI Imaging	(4) WR125

ABSTRACT

Signs of Polycyclic Aromatic Hydrocarbons (PAHs) are everywhere in the interstellar medium (ISM). PAHs account for almost 15% of the interstellar carbon in the ISM and radiate up to 20% of the total infrared power emitted by the Milky Way and star-forming galaxies. Not only are

PAHs thermal regulators of the ISM, but they also trace star formation and the chemical enrichment of galaxies. Despite their important role, the origins of PAHs are still uncertain. Carbon-rich Wolf-Rayet (WC) binaries have recently emerged as a potential source of PAHs. JWST has since thrust these systems into the spotlight, where spectroscopic and imaging observations revealed prominent PAH emission from nested circumstellar dust shells around the colliding-wind WC binary WR 140. Further JWST imaging has revealed nested circumstellar shells around all the dusty WC binaries it has observed. However, the abundance of PAHs in the dust shells formed by the systems other than WR 140 is uncertain due to the lack of spatially resolved spectroscopic observations, which are needed to confirm WC binaries as prolific PAH producers. To address this, we request 21.8 hours with MIRI to obtain spectroscopic and imaging observations of the newly resolved dust shells from the four WC binaries: WR 125, WR 112, WR 137, and WR 48a. The proposed observations would enable a crucial test on PAH abundance from WC binaries and allow us to quantify their PAH production efficiency from the measured fraction of PAH vs. dust mass. Based on their PAH production efficiency, the key science goal of this program is to determine whether WC binaries are significant sources of PAHs in the ISM of galaxies.

OBSERVING DESCRIPTION

We propose a total of 21.8 hr of JWST observations using the MIRI MRS and the MIRI Imager with the F1500W filter on the circumstellar shells of four dust-forming WC binaries: WR 125, WR 112, WR 137, and WR 48a. Motivated by previous JWST results from WR 140, the goal of the observations is to reveal and quantify the abundance of PAHs in the dust shells of our four targets to determine whether or not WC binaries can be significant sources of PAHs.

For each target, we request two MRS pointings to capture dust shells formed by the WC binaries in addition to a dedicated sky background as recommended by the JWST User Documentation. Given that PAH features are prominent around 6-12 microns, we set our MRS exposure times based on Cycle 2 MIRI imaging of our four proposed targets at 7.7 microns. We use the latest version of the ETC (v5.0) to determine the MRS exposure time required to achieve a signal-to-noise ratio (SNR) greater than 20 at 8 microns, which is needed to perform robust PAH and dust model fitting for measure PAH and dust masses. The science exposure times for the two offset pointings of each target is listed in the table above and utilize the FASTR1 readout pattern. The exposure time for the dedicated sky background observation is determined by the longer science exposure time. For each MRS observations, we request a 4-point dither pattern optimized for an extended source. No target acquisition is requested given the extended nature of the circumstellar dust shells.

Given the partial coverage of the dust shells from the proposed MRS observations, Imaging of our four targets with the standard 4-pt dither pattern using the F1500W filter is requested to confirm the full morphology of the dust shells. We note that the dust shells from our targets will have changed by $>0.2''$ in Cycle 5 from their initial imaging observations in 2024. We request the same exposure time (1432 s) as the Cycle 2 observations

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that revealed the dust shells around the four targets. The time requested for the Imager observations (including overheads) is only 20% of the total proposed observing time.

Proposal 10337 - Targets - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	WR137	RA: 20 14 31.7667 (303.6323613d) Dec: +36 39 39.60 (36.66100d) Equinox: J2000	Proper Motion RA: -2.681 mas/yr Proper Motion Dec: -5.739000084759027 mas/yr Parallax: 4.898E-4" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[WC stars, Wolf-Rayet stars]				
	(2)	WR112	RA: 18 16 33.4909 (274.1395454d) Dec: -18 58 42.34 (-18.97843d) Equinox: J2000	Proper Motion RA: -0.8760000000000001 mas/yr Proper Motion Dec: -3.7199999496806413 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[WC stars, Wolf-Rayet stars]				
	(3)	WR48a	RA: 13 12 39.6053 (198.1650221d) Dec: -62 42 55.94 (-62.71554d) Equinox: J2000	Proper Motion RA: -7.014 mas/yr Proper Motion Dec: -0.5999999984851456 mas/yr Parallax: 1.933E-4" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[WC stars, Wolf-Rayet stars]				
	(4)	WR125	RA: 19 28 15.6143 (292.0650596d) Dec: +19 33 21.53 (19.55598d) Equinox: J2000	Proper Motion RA: -3.5210000000000004 mas/yr Proper Motion Dec: -5.10300005771569 mas/yr Parallax: 1.3189999999999998E-4" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[WC stars, Wolf-Rayet stars]				
	(5)	WR48a-OFFSET1	RA: 13 12 40.5207 (198.1688362d) Dec: -62 42 56.72 (-62.71576d) Equinox: J2000	Parallax: 1.933E-4" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=ISM Description=[Dust nebulae, Stellar winds]					

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(6)	WR48a-OFFSET2	RA: 13 12 40.9642 (198.1706842d) Dec: -62 42 57.19 (-62.71589d) Equinox: J2000	Parallax: 1.933E-4" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=ISM Description=[Dust nebulae, Stellar winds]			
(7)	WR48a-Bkgd	RA: 13 12 44.1549 (198.1839787d) Dec: -62 43 27.66 (-62.72435d) Equinox: J2000	Parallax: 1.933E-4" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Telescope/sky background]			
(8)	WR137-OFFSET1	RA: 20 14 31.7506 (303.6322942d) Dec: +36 39 39.79 (36.66105d) Equinox: J2000	Parallax: 4.898E-4" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=ISM Description=[Dust nebulae, Stellar winds]			
(9)	WR137-OFFSET2	RA: 20 14 31.4570 (303.6310708d) Dec: +36 39 40.90 (36.66136d) Equinox: J2000	Parallax: 4.898E-4" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=ISM Description=[Dust nebulae, Stellar winds]			
(10)	WR137-Bkgd	RA: 20 14 30.4257 (303.6267737d) Dec: +36 38 58.07 (36.64946d) Equinox: J2000	Parallax: 4.898E-4" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Telescope/sky background]			
(11)	WR112-OFFSET1	RA: 18 16 33.0287 (274.1376196d) Dec: -18 58 47.30 (-18.97981d) Equinox: J2000	Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=ISM Description=[Dust nebulae, Stellar winds]			

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(12)	WR112-OFFSET2	RA: 18 16 32.8775 (274.1369896d) Dec: -18 58 48.18 (-18.98005d) Equinox: J2000	Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=ISM Description=[Dust nebulae, Stellar winds]			
(13)	WR112-Bkgd	RA: 18 16 29.6498 (274.1235408d) Dec: -18 58 20.04 (-18.97223d) Equinox: J2000	Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Telescope/sky background]			
(14)	WR125-OFFSET1	RA: 19 28 15.6444 (292.0651850d) Dec: +19 33 23.35 (19.55649d) Equinox: J2000	Parallax: 1.31899999999998E-4" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=ISM Description=[Dust nebulae, Stellar winds]			
(15)	WR125-OFFSET2	RA: 19 28 15.7757 (292.0657321d) Dec: +19 33 26.96 (19.55749d) Equinox: J2000	Parallax: 1.31899999999998E-4" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=ISM Description=[Dust nebulae, Stellar winds]			
(16)	WR125-Bkgd	RA: 19 28 16.0902 (292.0670425d) Dec: +19 32 41.00 (19.54472d) Equinox: J2000	Parallax: 1.31899999999998E-4" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Telescope/sky background]			

Proposal 10337 - Observation 1 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 1: WR48a-O1												Tue Apr 21 21:00:52 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(WR48a-O1 (Obs 1)) Warning (Form): Imager Filter overlap.												
	(WR48a-O1 (Obs 1)) Warning (Form): Imager Filter overlap.												
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(5)	WR48a-OFFSET1	RA: 13 12 40.5207 (198.1688362d)					Parallax: 1.933E-4"					
			Dec: -62 42 56.72 (-62.71576d)					Epoch of Position: 2000					
			Equinox: J2000										
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=ISM												
	Description=[Dust nebulae, Stellar winds]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	
	1	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
	1	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	
	2		IMAGER	F1500W	FASTR1	30	1	1	Dither 1	4	4	333.005	
	2	MEDIUM(B)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
	2	MEDIUM(B)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	
	3		IMAGER	F2100W	FASTR1	30	1	1	Dither 1	4	4	333.005	
	3	LONG(C)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
	3	LONG(C)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	

Proposal 10337 - Observation 1 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Special Requirements	Group Observations 1, 2, 4, 5, Non-interruptible
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Proposal 10337 - Observation 2 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 2: WR48a-O2												Tue Apr 21 21:00:52 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(WR48a-O2 (Obs 2)) Warning (Form): Imager Filter overlap.													
	(WR48a-O2 (Obs 2)) Warning (Form): Imager Filter overlap.													
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous				
	(6)	WR48a-OFFSET2	RA: 13 12 40.9642 (198.1706842d) Dec: -62 42 57.19 (-62.71589d) Equinox: J2000				Parallax: 1.933E-4" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=ISM Description=[Dust nebulae, Stellar winds]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction			
		All MRS				YES			FULL		Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2		IMAGER	F1500W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3		IMAGER	F2100W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		

Special Requirements	Group Observations 1, 2, 4, 5, Non-interruptible
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Proposal 10337 - Observation 5 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 5: WR48a-Bkgd												Tue Apr 21 21:00:52 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(WR48a-Bkgd (Obs 5)) Warning (Form): Imager Filter overlap.													
	(WR48a-Bkgd (Obs 5)) Warning (Form): Imager Filter overlap.													
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous				
	(7)	WR48a-Bkgd	RA: 13 12 44.1549 (198.1839787d) Dec: -62 43 27.66 (-62.72435d) Equinox: J2000				Parallax: 1.933E-4" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction			
		All MRS				YES			FULL		Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				BACKGROUND				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2		IMAGER	F1500W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3		IMAGER	F2100W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		

Special Requirements	Group Observations 1, 2, 4, 5, Non-interruptible
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Proposal 10337 - Observation 4 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 4: WR48a-Imaging										Tue Apr 21 21:00:52 GMT 2026
	Diagnostic Status: Warning										
Observing Template: MIRI Imaging											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	WR48a	RA: 13 12 39.6053 (198.1650221d) Dec: -62 42 55.94 (-62.71554d) Equinox: J2000			Proper Motion RA: -7.014 mas/yr Proper Motion Dec: -0.5999999984851456 mas/yr Parallax: 1.933E-4" Epoch of Position: 2000					
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
Category=Star											
Description=[WC stars, Wolf-Rayet stars]											
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				5	3	EXTENDED SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F1500W	FASTR1	43	1	1	Dither 1	12	12	1431.921	
Special Requirements	Group Observations 1, 2, 4, 5, Non-interruptible										

Proposal 10337 - Observation 6 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 6: WR137-O1												Tue Apr 21 21:00:52 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(WR137-O1 (Obs 6)) Warning (Form): Imager Filter overlap.													
	(WR137-O1 (Obs 6)) Warning (Form): Imager Filter overlap.													
	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous				
	(8)	WR137-OFFSET1	RA: 20 14 31.7506 (303.6322942d)				Parallax: 4.898E-4"							
			Dec: +36 39 39.79 (36.66105d)				Epoch of Position: 2000							
			Equinox: J2000											
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=ISM													
	Description=[Dust nebulae, Stellar winds]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction			
		All MRS				YES			FULL		Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2		IMAGER	F1500W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3		IMAGER	F2100W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		

Special Requirements	Group Observations 6, 7, 8, 9, Non-interruptible
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Proposal 10337 - Observation 7 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 7: WR137-O2												Tue Apr 21 21:00:52 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(WR137-O2 (Obs 7)) Warning (Form): Imager Filter overlap.													
	(WR137-O2 (Obs 7)) Warning (Form): Imager Filter overlap.													
	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(9)	WR137-OFFSET2	RA: 20 14 31.4570 (303.6310708d)				Parallax: 4.898E-4"							
			Dec: +36 39 40.90 (36.66136d)				Epoch of Position: 2000							
			Equinox: J2000											
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=ISM													
	Description=[Dust nebulae, Stellar winds]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
		All MRS				YES				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F770W	FASTR1	60	1	1	Dither 1	4	4	666.01		
	1	SHORT(A)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01		
	1	SHORT(A)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01		
	2		IMAGER	F1500W	FASTR1	60	1	1	Dither 1	4	4	666.01		
	2	MEDIUM(B)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01		
	2	MEDIUM(B)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01		
	3		IMAGER	F2100W	FASTR1	60	1	1	Dither 1	4	4	666.01		
	3	LONG(C)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01		
	3	LONG(C)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01		

Proposal 10337 - Observation 7 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Special Requirements	Group Observations 6, 7, 8, 9, Non-interruptible
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Proposal 10337 - Observation 8 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 8: WR137-Bkgd												Tue Apr 21 21:00:52 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(WR137-Bkgd (Obs 8)) Warning (Form): Imager Filter overlap.													
	(WR137-Bkgd (Obs 8)) Warning (Form): Imager Filter overlap.													
	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(10)	WR137-Bkgd	RA: 20 14 30.4257 (303.6267737d)				Parallax: 4.898E-4"							
			Dec: +36 38 58.07 (36.64946d)				Epoch of Position: 2000							
			Equinox: J2000											
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Calibration													
	Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
		All MRS				YES				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				BACKGROUND				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F770W	FASTR1	60	1	1	Dither 1	4	4	666.01		
	1	SHORT(A)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01		
	1	SHORT(A)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01		
	2		IMAGER	F1500W	FASTR1	60	1	1	Dither 1	4	4	666.01		
	2	MEDIUM(B)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01		
	2	MEDIUM(B)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01		
	3		IMAGER	F2100W	FASTR1	60	1	1	Dither 1	4	4	666.01		
	3	LONG(C)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01		
	3	LONG(C)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01		

Special Requirements	Group Observations 6, 7, 8, 9, Non-interruptible
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Proposal 10337 - Observation 9 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 9: WR137-Imaging										Tue Apr 21 21:00:52 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	WR137	RA: 20 14 31.7667 (303.6323613d)			Proper Motion RA: -2.681 mas/yr					
			Dec: +36 39 39.60 (36.66100d)			Proper Motion Dec: -5.739000084759027 mas/yr					
			Equinox: J2000			Parallax: 4.898E-4"					
						Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star										
	Description=[WC stars, Wolf-Rayet stars]										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				5	3	EXTENDED SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F1500W	FASTR1	43	1	1	Dither 1	12	12	1431.921	
Special Requirements	Group Observations 6, 7, 8, 9, Non-interruptible										

Proposal 10337 - Observation 11 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 11: WR112-O1													Tue Apr 21 21:00:52 GMT 2026
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(WR112-O1 (Obs 11)) Warning (Form): Imager Filter overlap. (WR112-O1 (Obs 11)) Warning (Form): Imager Filter overlap. (Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous			
	(11)	WR112-OFFSET1	RA: 18 16 33.0287 (274.1376196d) Dec: -18 58 47.30 (-18.97981d) Equinox: J2000					Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=ISM Description=[Dust nebulae, Stellar winds]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction			
		All MRS				YES			FULL		Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005		
	1	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	1	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		
	2		IMAGER	F1500W	FASTR1	30	1	1	Dither 1	4	4	333.005		
	2	MEDIUM(B)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	2	MEDIUM(B)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		
	3		IMAGER	F2100W	FASTR1	30	1	1	Dither 1	4	4	333.005		
	3	LONG(C)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	3	LONG(C)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		

Special Requirements	Group Observations 11, 12, 13, 14, Non-interruptible
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Proposal 10337 - Observation 12 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 12: WR112-O2												Tue Apr 21 21:00:52 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(WR112-O2 (Obs 12)) Warning (Form): Imager Filter overlap.													
	(WR112-O2 (Obs 12)) Warning (Form): Imager Filter overlap.													
	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(12)	WR112-OFFSET2	RA: 18 16 32.8775 (274.1369896d)				Epoch of Position: 2000							
			Dec: -18 58 48.18 (-18.98005d)											
			Equinox: J2000											
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=ISM													
	Description=[Dust nebulae, Stellar winds]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
		All MRS				YES				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2		IMAGER	F1500W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3		IMAGER	F2100W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		

Special Requirements	Group Observations 11, 12, 13, 14, Non-interruptible
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Proposal 10337 - Observation 13 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 13: WR112-Bkgd												Tue Apr 21 21:00:52 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(WR112-Bkgd (Obs 13)) Warning (Form): Imager Filter overlap.													
	(WR112-Bkgd (Obs 13)) Warning (Form): Imager Filter overlap.													
	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(13)	WR112-Bkgd	RA: 18 16 29.6498 (274.1235408d) Dec: -18 58 20.04 (-18.97223d) Equinox: J2000				Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
		All MRS				YES				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				BACKGROUND				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2		IMAGER	F1500W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3		IMAGER	F2100W	FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008		
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008		

Special Requirements	Group Observations 11, 12, 13, 14, Non-interruptible
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Proposal 10337 - Observation 14 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 14: WR112-Imaging										Tue Apr 21 21:00:52 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	WR112	RA: 18 16 33.4909 (274.1395454d) Dec: -18 58 42.34 (-18.97843d) Equinox: J2000			Proper Motion RA: -0.8760000000000001 mas/yr Proper Motion Dec: -3.7199999496806413 mas/yr Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star Description=[WC stars, Wolf-Rayet stars]										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				5	3	EXTENDED SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F1500W	FASTR1	43	1	1	Dither 1	12	12	1431.921	
Special Requirements	Group Observations 11, 12, 13, 14, Non-interruptible										

Proposal 10337 - Observation 16 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 16: WR125-O1												Tue Apr 21 21:00:52 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(WR125-O1 (Obs 16)) Warning (Form): Imager Filter overlap.													
	(WR125-O1 (Obs 16)) Warning (Form): Imager Filter overlap.													
	(Visit 16:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(14)	WR125-OFFSET1	RA: 19 28 15.6444 (292.0651850d)				Parallax: 1.318999999999998E-4"							
			Dec: +19 33 23.35 (19.55649d)				Epoch of Position: 2000							
			Equinox: J2000											
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=ISM													
	Description=[Dust nebulae, Stellar winds]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
		All MRS				YES				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005		
	1	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	1	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		
	2		IMAGER	F1500W	FASTR1	30	1	1	Dither 1	4	4	333.005		
	2	MEDIUM(B)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	2	MEDIUM(B)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		
	3		IMAGER	F2100W	FASTR1	30	1	1	Dither 1	4	4	333.005		
	3	LONG(C)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005		
	3	LONG(C)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005		

Special Requirements	Group Observations 16, 17, 18, 19, Non-interruptible
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Proposal 10337 - Observation 17 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 17: WR125-O2												Tue Apr 21 21:00:52 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(WR125-O2 (Obs 17)) Warning (Form): Imager Filter overlap.													
	(WR125-O2 (Obs 17)) Warning (Form): Imager Filter overlap.													
	(Visit 17:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(15)	WR125-OFFSET2	RA: 19 28 15.7757 (292.0657321d)				Parallax: 1.318999999999998E-4"							
			Dec: +19 33 26.96 (19.55749d)				Epoch of Position: 2000							
			Equinox: J2000											
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
	Category=ISM													
	Description=[Dust nebulae, Stellar winds]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				YES			FULL			Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F770W	FASTR1	60	1	1	Dither 1	4	4	666.01		
	1	SHORT(A)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01		
	1	SHORT(A)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01		
	2		IMAGER	F1500W	FASTR1	60	1	1	Dither 1	4	4	666.01		
	2	MEDIUM(B)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01		
	2	MEDIUM(B)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01		
	3		IMAGER	F2100W	FASTR1	60	1	1	Dither 1	4	4	666.01		
	3	LONG(C)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01		
	3	LONG(C)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01		

Special Requirements	Group Observations 16, 17, 18, 19, Non-interruptible
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Proposal 10337 - Observation 18 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 18: WR125-Bkgd												Tue Apr 21 21:00:52 GMT 2026
	Diagnostic Status: Warning												
Observing Template: MIRI Medium Resolution Spectroscopy													
Diagnostics	(WR125-Bkgd (Obs 18)) Warning (Form): Imager Filter overlap.												
	(WR125-Bkgd (Obs 18)) Warning (Form): Imager Filter overlap.												
(Visit 18:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(16)	WR125-Bkgd	RA: 19 28 16.0902 (292.0670425d)					Parallax: 1.318999999999998E-4"					
			Dec: +19 32 41.00 (19.54472d)					Epoch of Position: 2000					
			Equinox: J2000										
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
Category=Calibration													
Description=[Telescope/sky background]													
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					BACKGROUND			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	60	1	1	Dither 1	4	4	666.01	
	1	SHORT(A)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01	
	1	SHORT(A)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01	
	2		IMAGER	F1500W	FASTR1	60	1	1	Dither 1	4	4	666.01	
	2	MEDIUM(B)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01	
	2	MEDIUM(B)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01	
	3		IMAGER	F2100W	FASTR1	60	1	1	Dither 1	4	4	666.01	
	3	LONG(C)	MRSLONG		FASTR1	60	1	1	Dither 1	4	4	666.01	
	3	LONG(C)	MRSSHORT		FASTR1	60	1	1	Dither 1	4	4	666.01	

Proposal 10337 - Observation 18 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Special Requirements	Group Observations 16, 17, 18, 19, Non-interruptible
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Proposal 10337 - Observation 19 - Dusting for PAH-prints from Circumstellar Dust Shells around Wolf-Rayet Binaries

Observation	Proposal 10337, Observation 19: WR125-Imaging Diagnostic Status: Warning Observing Template: MIRI Imaging										Tue Apr 21 21:00:52 GMT 2026
Diagnostics	(Visit 19:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	WR125	RA: 19 28 15.6143 (292.0650596d) Dec: +19 33 21.53 (19.55598d) Equinox: J2000			Proper Motion RA: -3.5210000000000004 mas/yr Proper Motion Dec: -5.10300005771569 mas/yr Parallax: 1.3189999999999998E-4" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star Description=[WC stars, Wolf-Rayet stars]										
Template	Subarray FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				5	3	EXTENDED SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F1500W	FASTR1	43	1	1	Dither 1	12	12	1431.921	
Special Requirements	Group Observations 16, 17, 18, 19, Non-interruptible										